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Inventor(s)	Suzuki; Hiroyuki et al.

Control apparatus, system, vehicle, and medical examination support method

Abstract

A control apparatus includes a controller configured to acquire driving data regarding driving of a vehicle in which a medical examination can be performed, determine a driving condition of the vehicle based on the acquired driving data, and adjust implementation timing of at least one examination item included in the medical examination according to the determined driving condition.

Inventors: Suzuki; Hiroyuki (Miyoshi, JP), Kanemitsu; Yoshinori (Tachikawa, JP), Taniguchi; Hirohiko (Yokohama, JP), Ota; Hirona (Seto, JP), Tatsumoto; Yuki (Seto, JP), Uno; Keiichi (Aichi-ken, JP)

Applicant: TOYOTA JIDOSHA KABUSHIKI KAISHA (Toyota, JP)

Family ID: 87070544

Assignee: TOYOTA JIDOSHA KABUSHIKI KAISHA (Toyota, JP)

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Primary Examiner: Sorey; Robert A

Attorney, Agent or Firm: Dickinson Wright, PLLC

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

(1) This application claims priority to Japanese Patent Application No. 2022-002629 filed on Jan. 11, 2022, the entire contents of which are incorporated herein by reference.

TECHNICAL FIELD

(2) The present disclosure relates to a control apparatus, a system, a vehicle, and a medical examination support method.

BACKGROUND

(3) Patent Literature (PTL) 1 discloses a vehicle in which medical examinations can be performed.

CITATION LIST

Patent Literature

(4) PTL 1: JP 2021-022332 A

SUMMARY

(5) When medical examinations are performed in vehicles, some examination items may cause

danger due to phenomena such as vibration of the vehicles.

(6) It would be helpful to improve safety when medical examinations are performed in vehicles.

(7) A control apparatus according to the present disclosure includes a controller configured to: acquire driving data regarding driving of a vehicle in which a medical examination can be performed; determine a driving condition of the vehicle based on the acquired driving data; and adjust implementation timing of at least one examination item included in the medical examination according to the determined driving condition.

(8) A medical examination support method according to the present disclosure includes: acquiring, by a controller, driving data regarding driving of a vehicle in which a medical examination can be performed; determining, by the controller, a driving condition of the vehicle based on the acquired driving data; and adjusting, by the controller, implementation timing of at least one examination item included in the medical examination according to the determined driving condition.

(9) According to the present disclosure, safety is improved when medical examinations are performed in vehicles.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) In the accompanying drawings:

(2) FIG. 1 is a diagram illustrating a configuration of a system according to an embodiment of the present disclosure;

(3) FIG. 2 is a block diagram illustrating a configuration of a control apparatus according to the embodiment of the present disclosure;

(4) FIG. 3 is a flowchart illustrating operations of the control apparatus according to the embodiment of the present disclosure;

(5) FIG. 4 is a flowchart illustrating a variation of the operations of the control apparatus according to the embodiment of the present disclosure;

(6) FIG. 5 is a flowchart illustrating another variation of the operations of the control apparatus according to the embodiment of the present disclosure; and

(7) FIG. 6 is a flowchart illustrating yet another variation of the operations of the control apparatus according to the embodiment of the present disclosure.

DETAILED DESCRIPTION

(8) An embodiment of the present disclosure will be described below, with reference to the drawings.

(9) In the drawings, the same or corresponding portions are denoted by the same reference numerals. In the descriptions of the present embodiment, detailed descriptions of the same or corresponding portions are omitted or simplified, as appropriate.

(10) A configuration of a system **10** according to the present embodiment will be described with reference to FIG. 1.

(11) The system **10** according to the present embodiment includes a control apparatus **20** and a vehicle **30** in which medical examinations can be performed. The control apparatus **20** can communicate with the vehicle **30** via a network **40**.

(12) The control apparatus **20** is installed in a facility such as a data center. The control apparatus **20** is a computer such as a server that belongs to a cloud computing system or another type of computing system.

(13) The vehicle **30** is equipped with medical examination equipment so that a medical examination can be performed in the vehicle **30**. The medical examination includes, as one or more examination items, height measurement, weight measurement, abdominal circumference measurement, vision test, hearing test, blood pressure test, urine collection, blood collection, chest

radiography, stomach radiography, ultrasonography, electrocardiography, CT test, MRI test, mammography, osteoporosis test, medical interview, or any combination of the above. The term “CT” is an abbreviation of computed tomography. The term “MRI” is an abbreviation of magnetic resonance imaging. As the medical examination equipment, equipment corresponding to each of the examination items is mounted. In other words, a height meter, a weight meter, vision testing equipment, hearing testing equipment, a blood pressure meter, urinalysis equipment, blood testing equipment, X-ray imaging equipment, ultrasound equipment, an electrocardiograph, CT testing equipment, MRI testing equipment, or any combination of the above may be mounted. As ancillary equipment, a bed on which a user **11** having the medical examination may lie, or a chair on which the user **11** may sit. The vehicle **30** is, for example, any type of automobile such as a gasoline vehicle, a diesel vehicle, a hydrogen vehicle, an HEV, a PHEV, a BEV, or an FCEV. The term “HEV” is an abbreviation of hybrid electric vehicle. The term “PHEV” is an abbreviation of plug-in hybrid electric vehicle. The term “BEV” is an abbreviation of battery electric vehicle. The term “FCEV” is an abbreviation of fuel cell electric vehicle. The vehicle **30**, which is an AV in the present embodiment, may be driven by a driver, or the driving may be automated at any level. The term “AV” is an abbreviation of autonomous vehicle. The automation level is, for example, any one of Level 1 to Level 5 according to the level classification defined by SAE. The name “SAE” is an abbreviation of Society of Automotive Engineers. The vehicle **30** may be a MaaS-dedicated vehicle. The term “MaaS” is an abbreviation of Mobility as a Service.

(14) The network **40** includes the Internet, at least one WAN, at least one MAN, or any combination thereof. The term “WAN” is an abbreviation of wide area network. The term “MAN” is an abbreviation of metropolitan area network. The network **40** may include at least one wireless network, at least one optical network, or any combination thereof. The wireless network is, for example, an ad hoc network, a cellular network, a wireless LAN, a satellite communication network, or a terrestrial microwave network. The term “LAN” is an abbreviation of local area network.

(15) An outline of the present embodiment will be described with reference to FIG. 1.

(16) The control apparatus **20** acquires driving data Dd regarding driving of the vehicle **30**. The control apparatus **20** determines a driving condition Cd of the vehicle **30** based on the acquired driving data Dd. The control apparatus **20** adjusts, according to the determined driving condition Cd, implementation timing Ti of at least one examination item included in the medical examination to be performed in the vehicle **30**.

(17) According to the present embodiment, in the case of performing the medical examination in the vehicle **30**, the medical examination can be performed in conjunction with movement of the vehicle **30**, such as postponing a blood test using an injection needle when the vehicle **30** is moving at high speed, starting, braking, or traveling on a rough road. As a result, safety is improved when the medical examination is performed in the vehicle **30**.

(18) A configuration of the control apparatus **20** according to the present embodiment will be described with reference to FIG. 2.

(19) The control apparatus **20** includes a controller **21**, a memory **22**, and a communication interface **23**.

(20) The controller **21** includes at least one processor, at least one programmable circuit, at least one dedicated circuit, or any combination thereof. The processor is a general purpose processor such as a CPU or a GPU, or a dedicated processor that is dedicated to specific processing. The term “CPU” is an abbreviation of central processing unit. The term “GPU” is an abbreviation of graphics processing unit. The programmable circuit is, for example, an FPGA. The term “FPGA” is an abbreviation of field-programmable gate array. The dedicated circuit is, for example, an ASIC. The term “ASIC” is an abbreviation of application specific integrated circuit. The controller **21** executes processes related to operations of the control apparatus **20** while controlling components of the control apparatus **20**.

(21) The memory **22** includes at least one semiconductor memory, at least one magnetic memory, at least one optical memory, or any combination thereof. The semiconductor memory is, for example, RAM or ROM. The term “RAM” is an abbreviation of random access memory. The term “ROM” is an abbreviation of read only memory. The RAM is, for example, SRAM or DRAM. The term “SRAM” is an abbreviation of static random access memory. The term “DRAM” is an abbreviation of dynamic random access memory. The ROM is, for example, EEPROM. The term “EEPROM” is an abbreviation of electrically erasable programmable read only memory. The memory **22** functions as, for example, a main memory, an auxiliary memory, or a cache memory. The memory **22** stores data to be used for the operations of the control apparatus **20** and data obtained by the operations of the control apparatus **20**.

(22) The communication interface **23** includes at least one interface for communication. The interface for communication is, for example, a LAN interface. The communication interface **23** communicates with the vehicle **30**. The communication interface **23** receives data to be used for the operations of the control apparatus **20**, and transmits data obtained by the operations of the control apparatus **20**.

(23) The functions of the control apparatus **20** are realized by execution of a program according to the present embodiment by a processor serving as the controller **21**. That is, the functions of the control apparatus **20** are realized by software. The program causes a computer to execute the operations of the control apparatus **20**, thereby causing the computer to function as the control apparatus **20**. That is, the computer executes the operations of the control apparatus **20** in accordance with the program to thereby function as the control apparatus **20**.

(24) The program can be stored on a non-transitory computer readable medium. The non-transitory computer readable medium is, for example, flash memory, a magnetic recording device, an optical disc, a magneto-optical recording medium, or ROM. The program is distributed, for example, by selling, transferring, or lending a portable medium such as an SD card, a DVD, or a CD-ROM on which the program is stored. The term “SD” is an abbreviation of Secure Digital. The term “DVD” is an abbreviation of digital versatile disc. The term “CD-ROM” is an abbreviation of compact disc read only memory. The program may be distributed by storing the program in a storage of a server and transferring the program from the server to another computer. The program may be provided as a program product.

(25) For example, the computer temporarily stores, in a main memory, a program stored in a portable medium or a program transferred from a server. Then, the computer reads the program stored in the main memory using a processor, and executes processes in accordance with the read program using the processor. The computer may read a program directly from the portable medium, and execute processes in accordance with the program. The computer may, each time a program is transferred from the server to the computer, sequentially execute processes in accordance with the received program. Instead of transferring a program from the server to the computer, processes may be executed by a so-called ASP type service that realizes functions only by execution instructions and result acquisitions. The term “ASP” is an abbreviation of application service provider.

Programs encompass information that is to be used for processing by an electronic computer and is thus equivalent to a program. For example, data that is not a direct command to a computer but has a property that regulates processing of the computer is “equivalent to a program” in this context.

(26) Some or all of the functions of the control apparatus **20** may be realized by a programmable circuit or a dedicated circuit serving as the controller **21**. That is, some or all of the functions of the control apparatus **20** may be realized by hardware.

(27) Operations of the control apparatus **20** according to the present embodiment will be described with reference to FIG. **3**. These operations correspond to a medical examination support method according to the present embodiment.

(28) In step **S101**, the controller **21** of the control apparatus **20** acquires driving data **Dd** regarding driving of the vehicle **30**. The driving data **Dd** is specifically acquired in any one of the following

two procedures.

(29) In the first procedure, the vehicle **30** transmits positioning results obtained by a GNSS receiver mounted on the vehicle **30** to the control apparatus **20**, as position data D_p , two or more times via a communication interface of the vehicle **30**, which is compliant with a mobile communication standard such as LTE, the 4G standard, or the 5G standard. The term “LTE” is an abbreviation of Long Term Evolution. The term “4G” is an abbreviation of 4th generation. The term “5G” is an abbreviation of 5th generation. The term “GNSS” is an abbreviation of global navigation satellite system. GNSS is, for example, GPS, QZSS, BDS, GLONASS, or Galileo. The term “GPS” is an abbreviation of Global Positioning System. The term “QZSS” is an abbreviation of Quasi-Zenith Satellite System. QZSS satellites are called quasi-zenith satellites. The term “BDS” is an abbreviation of BeiDou Navigation Satellite System. The term “GLONASS” is an abbreviation of Global Navigation Satellite System. The controller **21** of the control apparatus **20** receives the position data D_p from the vehicle **30** via the communication interface **23**. Whenever the controller **21** receives the position data D_p , the controller **21** stores the received position data D_p in the memory **22**, thereby accumulating, as the driving data D_d , data D_{d1} indicating the position of the vehicle **30** in time series in the memory **22**. In other words, the controller **21** acquires the driving data D_d by receiving the data D_{d1} indicating the position of the vehicle **30** in time series from the vehicle **30** via the communication interface **23**.

(30) In the second procedure, the vehicle **30** transmits a sensing result obtained by a sensor such as a camera, LiDAR, or radar mounted on the vehicle **30**, as sensor data D_s , at least once to the control apparatus **20** via the communication interface of the vehicle **30**. The term “LiDAR” is an abbreviation of light detection and ranging. The controller **21** of the control apparatus **20** receives the sensor data D_s from the vehicle **30** via the communication interface **23**. Upon receiving the sensor data D_s , the controller **21** stores the received sensor data D_s in the memory **22**, thereby saving, as the driving data D_d , data D_{d2} indicating the sensing result obtained by the sensor mounted on the vehicle **30** in the memory **22**. In other words, the controller **21** acquires the driving data D_d by receiving the data D_{d2} indicating the sensing result obtained by the sensor mounted on the vehicle **30** from the vehicle **30** via the communication interface **23**.

(31) In step **S102**, the controller **21** of the control apparatus **20** determines a driving condition C_d of the vehicle **30** based on the driving data D_d acquired in step **S101**. Specifically, the controller **21** determines, as the driving condition C_d , whether the speed of the vehicle **30** exceeds a first threshold value $Th1$. Whether the speed of the vehicle **30** exceeds the first threshold value $Th1$ is specifically determined by the following procedure.

(32) When the data D_{d1} is acquired as the driving data D_d in step **S101**, the controller **21** of the control apparatus **20** analyzes a change in position indicated by the acquired data D_{d1} to determine whether the speed of the vehicle **30** exceeds the first threshold value $Th1$. For example, the controller **21** calculates the speed of the vehicle **30** by dividing a distance between the current position of the vehicle **30** and the immediately preceding position thereof, indicated by the acquired data D_{d1} , by a difference between corresponding two positioning times. The controller **21** then determines whether the calculated speed exceeds the first threshold value $Th1$.

(33) When the data D_{d2} is acquired as the driving data D_d in step **S101**, the controller **21** of the control apparatus **20** analyzes the sensing result indicated by the acquired data D_{d2} to determine whether the speed of the vehicle **30** exceeds the first threshold value $Th1$. For example, the controller **21** calculates the speed of the vehicle **30** by analyzing video, which corresponds to the acquired data D_{d2} , captured by the camera mounted on the vehicle **30**. The controller **21** then determines whether the calculated speed exceeds the first threshold value $Th1$. As a video analysis method, a known method can be used. Machine learning, such as deep learning, may be used.

(34) In the operations illustrated in FIG. 3, for example, the first threshold value $Th1$ is set to a value between 50 km/h and 80 km/h inclusive. Thus, it is possible to determine, as the driving condition C_d , whether the vehicle **30** is moving at high speed. Alternatively, the first threshold

value **Th1** may be set to 0 km/h. In such an example, it is possible to determine, as the driving condition **Cd**, whether the vehicle **30** is moving or stopped.

(35) When it is determined in step **S102** that the speed of the vehicle **30** exceeds the first threshold value **Th1**, the process in step **S103** is performed. When it is determined in step **S102** that the speed of vehicle **30** does not exceed the first threshold value **Th1**, the operations illustrated in FIG. 3 are ended.

(36) In step **S103**, the controller **21** of the control apparatus **20** predicts, based on the driving data **Dd** acquired in step **S101**, a time **Tm1** when the speed of the vehicle **30** comes to be equal to or less than the first threshold value **Th1**. The time **Tm1** is specifically predicted by the following procedure.

(37) When the data **Dd1** is acquired as the driving data **Dd** in step **S101**, the controller **21** of the control apparatus **20** receives road traffic information such as a speed limit, traffic congestion, or timing of changing a traffic light on a road beyond the current position of the vehicle **30** indicated by the acquired data **Dd1** from an external system via the communication interface **23**. The controller **21** predicts the time **Tm1** when the speed of the vehicle **30** comes to be equal to or less than the first threshold value **Th1** with reference to the received road traffic information.

(38) When the data **Dd2** is acquired as the driving data **Dd** in step **S101**, the controller **21** of the control apparatus **20** generates road traffic information such as a speed limit, traffic congestion, or timing of changing a traffic light on a road beyond the current position of the vehicle **30** by analyzing the video, which corresponds to the acquired data **Dd2**, captured by the camera mounted on the vehicle **30**. As a video analysis method, a known method can be used. Machine learning, such as deep learning, may be used. The controller **21** predicts the time **Tm1** when the speed of the vehicle **30** comes to be equal to or less than the first threshold value **Th1** with reference to the generated road traffic information.

(39) In step **S104**, the controller **21** of the control apparatus **20** adjusts, according to the driving condition **Cd** determined in step **S102**, implementation timing **Ti** of at least one examination item included in the medical examination to be performed in the vehicle **30**. Specifically, when it is determined in step **S102** that the speed of the vehicle **30** exceeds the first threshold value **Th1**, the controller **21** adjusts the implementation timing **Ti** by outputting request data **Dq** requesting to postpone performing the at least one examination item until the speed of the vehicle **30** comes to be equal to or less than the first threshold value **Th1**. The controller **21** adds data indicating the time **Tm1** predicted in step **S103** to the request data **Dq**. The request data **Dq** is specifically output by the following procedure.

(40) The controller **21** of the control apparatus **20** outputs the request data **Dq** to the vehicle **30**. That is, the controller **21** transmits the request data **Dq** to the vehicle **30** via the communication interface **23**. The vehicle **30** receives the request data **Dq** from the control apparatus **20** via the communication interface of the vehicle **30**. The vehicle **30** displays a text message corresponding to the received request data **Dq** on a display mounted on the vehicle **30**. Alternatively, the vehicle **30** may output an audio message corresponding to the received request data **Dq** from a speaker mounted on the vehicle **30**. The display is, for example, an LCD or an organic EL display. The term “LCD” is an abbreviation of liquid crystal display. The term “EL” is an abbreviation of electro luminescence. The text message to be displayed on the display may include the time **Tm1** indicated by the data added to the request data **Dq**. The message to be output from the speaker may also include the time **Tm1**.

(41) In one example, the request data **Dq** includes data requesting to postpone performing a test using an injection needle, such as a blood test, as the at least one examination item. The vehicle **30** displays, according to the request data **Dq**, a text message such as “Do not perform a blood test because the vehicle is moving at high speed” on the display. Alternatively, the vehicle **30** may output an audio message of similar contents from the speaker. When the time **Tm1** is X minutes after the current time, the vehicle **30** may display, on the display, a text message such as “Please

perform a blood test in X minutes because the vehicle is moving at high speed”. Alternatively, the vehicle **30** may output an audio message of similar contents from the speaker.

(42) According to this example, the examination using the injection needle can be postponed while the vehicle **30** is moving at high speed. As a result, safety is improved when the medical examination is performed in the vehicle **30**.

(43) In another example, the request data Dq includes data requesting to postpone performing a test to be performed after the user **11** once stands up, such as a test subsequent to a test performed on the user **11** sitting on a chair, as the at least one examination item. The vehicle **30** displays, according to the request data Dq, a text message such as “Do not let the patient stand up from the chair because the vehicle is moving at high speed” on the display. Alternatively, the vehicle **30** may output an audio message of similar contents from the speaker. When the time Tm1 is X minutes after the current time, the vehicle **30** may display, on the display, a text message such as “Please let the patient stand up from the chair in X minutes because the vehicle is moving at high speed”. Alternatively, the vehicle **30** may output an audio message of similar contents from the speaker.

(44) According to this example, it is possible to keep the user **11** from standing up while the vehicle **30** is moving at high speed, especially when the user **11** is an elderly person. As a result, safety is improved when the medical examination is performed in the vehicle **30**.

(45) The operations illustrated in FIG. 3 are repeated periodically or whenever a specific event occurs, such as a change in the driving condition of the vehicle **30**.

(46) In step S101, the controller **21** of the control apparatus **20** may acquire, as the driving data Dd, data Dd3 indicating the speed of the vehicle **30**. Specifically, the controller **21** may receive the data Dd3 indicating the speed of the vehicle **30** from the vehicle **30** via the communication interface **23**. In this case, in step S102, the controller **21** simply determines whether the speed indicated by the acquired data Dd3 exceeds the first threshold value Th1.

(47) The process in step S103 may be omitted. In other words, the request data Dq output in step S104 need not be added with data indicating the time Tm1 when the speed of the vehicle **30** comes to be equal to or less than the first threshold value Th1.

(48) In step S104, the controller **21** of the control apparatus **20** may output the request data Dq to a terminal apparatus of a medical staff, such as a nurse, performing the medical examination. In other words, the controller **21** may transmit the request data Dq to the terminal apparatus of the medical staff via the communication interface **23**. The terminal apparatus is, for example, a mobile device such as a mobile phone, a smartphone, or a tablet, or a PC. The term “PC” is an abbreviation of personal computer. The terminal apparatus receives the request data Dq from the control apparatus **20** via a communication interface of the terminal apparatus compliant with a mobile communication standard such as LTE, the 4G standard, or the 5G standard. The terminal apparatus displays, on a display of the terminal apparatus, a text message according to the received request data Dq. Alternatively, the terminal apparatus may output an audio message corresponding to the received request data Dq from a speaker of the terminal apparatus.

(49) A variation of the operations illustrated in FIG. 3 will be described with reference to FIG. 4. The process in step S201 is the same as the process in step S101 of FIG. 3, and thus a description thereof is omitted.

(50) In step S202, the controller **21** of the control apparatus **20** determines a driving condition Cd of the vehicle **30** based on the driving data Dd acquired in step S201. Specifically, the controller **21** determines, as the driving condition Cd, whether the acceleration of the vehicle **30** transitorily exceeds a second threshold value Th2. Whether the acceleration of the vehicle **30** transitorily exceeds the second threshold value Th2 is specifically determined by the following procedure.

(51) When the data Dd1 is acquired as the driving data Dd in step S201, the controller **21** of the control apparatus **20** analyzes a change in position indicated by the acquired data Dd1 to determine whether the acceleration of the vehicle **30** transitorily exceeds the second threshold value Th2. For example, the controller **21** determines whether the current position of the vehicle **30** and the

immediately preceding position thereof, indicated by the acquired data Dd1, are the same, that is, whether the vehicle 30 is stopped. When it is determined that the vehicle 30 is stopped, the controller 21 refers to map data Dm and determines whether there is a traffic light at the current position of the vehicle 30, in other words, whether the vehicle 30 is waiting for the traffic light to change. When it is determined that the vehicle 30 is waiting for the traffic light to change, the controller 21 determines that the acceleration of the vehicle 30 transitorily exceeds the second threshold value Th2, in other words, the vehicle 30 starts or in some cases accelerates rapidly when the traffic light changes. The map data Dm may be stored in advance in the memory 22 of the control apparatus 20, or may be accumulated in an external system such as an Internet-based GIS. The term “GIS” is an abbreviation of geographic information system.

(52) When the data Dd2 is acquired as the driving data Dd in step S201, the controller 21 of the control apparatus 20 analyzes the sensing result indicated by the acquired data Dd2 to determine whether the acceleration of the vehicle 30 transitorily exceeds the second threshold value Th2. For example, the controller 21 analyzes the video captured by the camera mounted on the vehicle 30, which corresponds to the acquired data Dd2, to determine whether the vehicle 30 is waiting at a traffic light to change. When it is determined that the vehicle 30 is waiting for the traffic light to change, the controller 21 determines that the acceleration of the vehicle 30 transitorily exceeds the second threshold value Th2, in other words, the vehicle 30 starts or in some cases accelerates rapidly when the traffic light changes. As a video analysis method, a known method can be used. Machine learning, such as deep learning, may be used.

(53) In the operations illustrated in FIG. 4, for example, the second threshold value Th2 is set to a value equal to or more than 8 km/h/s and less than 10 km/h/s. Therefore, whether the vehicle 30 is about to start can be determined as the driving condition Cd. Alternatively, the second threshold value Th2 may be set to a value equal to or more than 10 km/h/s. In such an example, whether the vehicle 30 is about to accelerate rapidly can be determined as the driving condition Cd.

(54) When it is determined in step S202 that the acceleration of the vehicle 30 transitorily exceeds the second threshold value Th2, the process in step S203 is performed. When it is determined in step S202 that the acceleration of the vehicle 30 does not exceed the second threshold value Th2, the operations illustrated in FIG. 4 are ended.

(55) In step S203, the controller 21 of the control apparatus 20 adjusts, according to the driving condition Cd determined in step S202, implementation timing Ti of at least one examination item included in the medical examination to be performed in the vehicle 30. Specifically, when it is determined in step S202 that the acceleration of the vehicle 30 transitorily exceeds the second threshold value Th2, the controller 21 adjusts the implementation timing Ti by outputting request data Dq requesting to postpone performing the at least one examination item until the acceleration of the vehicle 30 comes to be equal to or less than the second threshold value Th2 again. A specific procedure for outputting the request data Dq is the same as the procedure in step S104 of FIG. 3, so a description thereof is omitted.

(56) In one example, the request data Dq includes data requesting to postpone performing a test using an injection needle, such as a blood test, as the at least one examination item. The vehicle 30 displays, according to the request data Dq, a text message such as “Do not perform a blood test because the vehicle is about to start” on the display. Alternatively, the vehicle 30 may output an audio message of similar contents from the speaker.

(57) According to this example, the examination using the injection needle can be postponed while the vehicle 30 is starting. As a result, safety is improved when the medical examination is performed in the vehicle 30.

(58) In another example, the request data Dq includes data requesting to postpone performing a test to be performed after the user 11 once stands up, such as a test subsequent to a test performed on the user 11 sitting on a chair, as the at least one examination item. The vehicle 30 displays, according to the request data Dq, a text message such as “Do not let the patient stand up from the

chair because the vehicle is about to start” on the display. Alternatively, the vehicle **30** may output an audio message of similar contents from the speaker.

(59) According to this example, it is possible to keep the user **11** from standing up while the vehicle **30** is starting, especially when the user **11** is an elderly person. As a result, safety is improved when the medical examination is performed in the vehicle **30**.

(60) The operations illustrated in FIG. **4** are repeated periodically or whenever a specific event occurs, such as a change in the driving condition of the vehicle **30**.

(61) In step **S203**, as in step **S104** of FIG. **3**, the controller **21** of the control apparatus **20** may output the request data **Dq** to the terminal apparatus of the medical staff.

(62) Another variation of the operations illustrated in FIG. **3** will be described with reference to FIG. **5**. The process in step **S301** is the same as the process in step **S101** of FIG. **3**, and thus a description thereof is omitted.

(63) In step **S302**, the controller **21** of the control apparatus **20** determines a driving condition **Cd** of the vehicle **30** based on the driving data **Dd** acquired in step **S301**. Specifically, the controller **21** determines, as the driving condition **Cd**, whether the deceleration of the vehicle **30** transitorily exceeds a third threshold value **Th3**. Whether the deceleration of the vehicle **30** transitorily exceeds the third threshold value **Th3** is specifically determined by the following procedure.

(64) When the data **Dd1** is acquired as the driving data **Dd** in step **S301**, the controller **21** of the control apparatus **20** analyzes a change in position indicated by the acquired data **Dd1** to determine whether the deceleration of the vehicle **30** transitorily exceeds the third threshold value **Th3**. For example, the controller **21** determines whether the current position of the vehicle **30** and the immediately preceding position thereof, indicated by the acquired data **Dd1**, are different, that is, whether the vehicle **30** is moving. When it is determined that the vehicle **30** is moving, the controller **21** refers to map data **Dm** and determines whether there is a traffic light beyond the current position of the vehicle **30**. When it is determined that there is a traffic light beyond the current position of the vehicle **30**, the controller **21** receives road traffic information, such as timing of changing the traffic light, from an external system via the communication interface **23**. The controller **21** refers to the received road traffic information and determines whether the vehicle **30** stops or in some cases decelerates rapidly at the traffic light, in other words, whether the deceleration of the vehicle **30** transitorily exceeds the third threshold value **Th3**.

(65) When the data **Dd2** is acquired as the driving data **Dd** in step **S301**, the controller **21** of the control apparatus **20** analyzes the sensing result indicated by the acquired data **Dd2** to determine whether the deceleration of the vehicle **30** transitorily exceeds the third threshold value **Th3**. For example, the controller **21** analyzes the video captured by the camera mounted on the vehicle **30**, which corresponds to the acquired data **Dd2**, to determine whether the vehicle **30** is moving, there is a traffic light beyond the current position of the vehicle **30**, and the vehicle **30** stops or in some cases decelerates rapidly at the traffic light, in other words, whether the deceleration of the vehicle **30** transitorily exceeds the third threshold value **Th3**. As a video analysis method, a known method can be used. Machine learning, such as deep learning, may be used.

(66) In the operations illustrated in FIG. **5**, for example, the third threshold value **Th3** is set to a value equal to or more than 8 km/h/s and less than 10 km/h/s. Therefore, whether the vehicle **30** is about to stop can be determined as the driving condition **Cd**. Alternatively, the third threshold value **Th3** may be set to a value equal to or more than 10 km/h/s. In such an example, whether the vehicle **30** is about to decelerate rapidly can be determined as the driving condition **Cd**.

(67) When it is determined in step **S302** that the deceleration of the vehicle **30** transitorily exceeds the third threshold value **Th3**, the process in step **S303** is performed. When it is determined in step **S302** that the deceleration of the vehicle **30** does not exceed the third threshold value **Th3**, the operations illustrated in FIG. **5** are ended.

(68) In step **S303**, the controller **21** of the control apparatus **20** adjusts, according to the driving condition **Cd** determined in step **S302**, implementation timing **Ti** of at least one examination item

included in the medical examination to be performed in the vehicle **30**. Specifically, when it is determined in step **S302** that the deceleration of the vehicle **30** transitorily exceeds the third threshold value **Th3**, the controller **21** adjusts the implementation timing **Ti** by outputting request data **Dq** requesting to postpone performing the at least one examination item until the deceleration of the vehicle **30** comes to be equal to or less than the third threshold value **Th3** again. A specific procedure for outputting the request data **Dq** is the same as the procedure in step **S104** of FIG. **3**, so a description thereof is omitted.

(69) In one example, the request data **Dq** includes data requesting to postpone performing a test using an injection needle, such as a blood test, as the at least one examination item. The vehicle **30** displays, according to the request data **Dq**, a text message such as “Do not perform a blood test because the vehicle is about to stop” on the display. Alternatively, the vehicle **30** may output an audio message of similar contents from the speaker.

(70) According to this example, the examination using the injection needle can be postponed while the vehicle **30** is braking. As a result, safety is improved when the medical examination is performed in the vehicle **30**.

(71) In another example, the request data **Dq** includes data requesting to postpone performing a test to be performed after the user **11** once stands up, such as a test subsequent to a test performed on the user **11** sitting on a chair, as the at least one examination item. The vehicle **30** displays, according to the request data **Dq**, a text message such as “Do not let the patient stand up from the chair because the vehicle is about to stop” on the display. Alternatively, the vehicle **30** may output an audio message of similar contents from the speaker.

(72) According to this example, it is possible to keep the user **11** from standing up while the vehicle **30** is braking, especially when the user **11** is an elderly person. As a result, safety is improved when the medical examination is performed in the vehicle **30**.

(73) The operations illustrated in FIG. **5** are repeated periodically or whenever a specific event occurs, such as a change in the driving condition of the vehicle **30**.

(74) In step **S303**, as in step **S104** of FIG. **3**, the controller **21** of the control apparatus **20** may output the request data **Dq** to the terminal apparatus of the medical staff.

(75) Further another variation of the operations illustrated in FIG. **3** will be described with reference to FIG. **6**. The process in step **S401** is the same as the process in step **S101** of FIG. **3**, and thus a description thereof is omitted.

(76) In step **S402**, the controller **21** of the control apparatus **20** determines a driving condition **Cd** of the vehicle **30** based on driving data **Dd** acquired in step **S401**. Specifically, the controller **21** determines, as the driving condition **Cd**, a road surface condition **Rc** of a road on which the vehicle **30** travels. The road surface condition **Rc** is specifically determined by the following procedure.

(77) When the data **Dd1** is acquired as the driving data **Dd** in step **S401**, the controller **21** of the control apparatus **20** analyzes a change in position indicated by the acquired data **Dd1** and identifies a road on which the vehicle **30** travels, to determine the road surface condition **Rc** of the identified road. For example, the controller **21** determines whether the current position of the vehicle **30** and the immediately preceding position thereof, indicated by the acquired data **Dd1**, are different, that is, whether the vehicle **30** is moving. When it is determined that the vehicle **30** is moving, the controller **21** identifies a road beyond the current position of the vehicle **30** with reference to map data **Dm**. The controller **21** then determines the road surface condition **Rc** of the identified road with reference to information included in the map data **Dm** and information obtained from an external system.

(78) When the data **Dd2** is acquired as the driving data **Dd** in step **S401**, the controller **21** of the control apparatus **20** analyzes the sensing result indicated by the acquired data **Dd2** to determine the road surface condition **Rc** of the road on which the vehicle **30** travels. For example, the controller **21** analyzes point group data, which corresponds to the acquired data **Dd2**, obtained by LiDAR mounted on the vehicle **30**, to determine the road surface condition **Rc** of the road beyond

the current position of the vehicle **30**. As a point group data analysis method, a known method can be used. Machine learning, such as deep learning, may be used.

(79) In the operations illustrated in FIG. **6**, the road surface condition R_c is determined depending on whether the road meets a certain criterion, such as being paved.

(80) When the road surface condition R_c determined in step **S402** does not meet the criterion, the process in step **S403** is performed. When the road surface condition R_c determined in step **S402** meets the criterion, the operations illustrated in FIG. **6** are ended.

(81) In step **S403**, the controller **21** of the control apparatus **20** predicts, based on the driving data D_d acquired in step **S401**, a time T_{m4} when the vehicle **30** arrives at a road meeting the criterion. The time T_{m4} is specifically predicted by the following procedure.

(82) When the data D_{d1} is acquired as the driving data D_d in step **S401**, the controller **21** of the control apparatus **20** identifies a group of roads further ahead of the road beyond the current position of the vehicle **30** with reference to the map data D_m . The controller **21** identifies a road that meets the criterion from among the identified group of roads with reference to information included in the map data D_m or information obtained from an external system. The controller **21** receives, from an external system via the communication interface **23**, road traffic information such as a speed limit, traffic congestion, or timing of changing a traffic light on each road between the current position of the vehicle **30** indicated by the acquired data D_{d1} and the road meeting the criterion. The controller **21** predicts the time T_{m4} when the vehicle **30** arrives at the road meeting the criterion, with reference to the received road traffic information.

(83) When the data D_{d2} is acquired as the driving data D_d in step **S401**, the controller **21** of the control apparatus **20** identifies a road that is further ahead of the road beyond the current position of the vehicle **30** and that meets the criterion by analyzing the point group data, which corresponds to the acquired data D_{d2} , obtained by the LiDAR mounted on the vehicle **30**, and generates road traffic information such as a speed limit, traffic congestion, or timing of changing a traffic light on each road between the current position of the vehicle **30** and the road meeting the criterion. As a point group data analysis method, a known method can be used. Machine learning, such as deep learning, may be used. The controller **21** predicts the time T_{m4} when the vehicle **30** arrives at the road meeting the criterion, with reference to the generated road traffic information.

(84) In step **S404**, the controller **21** of the control apparatus **20** adjusts, according to the driving condition C_d determined in step **S402**, implementation timing T_i of at least one examination item included in the medical examination to be performed in the vehicle **30**. Specifically, when the road surface condition R_c determined in step **S402** does not meet the criterion, the controller **21** adjusts the implementation timing T_i by outputting request data D_q requesting to postpone performing the at least one examination item until the vehicle **30** arrives at the road meeting the criterion. The controller **21** adds data indicating the time T_{m4} predicted in step **S403** to the request data D_q . The request data D_q is specifically output by the following procedure.

(85) The controller **21** of the control apparatus **20** outputs the request data D_q to the vehicle **30**. That is, the controller **21** transmits the request data D_q to the vehicle **30** via the communication interface **23**. The vehicle **30** receives the request data D_q from the control apparatus **20** via the communication interface of the vehicle **30**. The vehicle **30** displays a text message corresponding to the received request data D_q on the display mounted on the vehicle **30**. Alternatively, the vehicle **30** may output an audio message corresponding to the received request data D_q from the speaker mounted on the vehicle **30**. The text message displayed on the display may include the time T_{m4} indicated by the data added to the request data D_q . The message to be output from the speaker may also include the time T_{m4} .

(86) In one example, the request data D_q includes data requesting to postpone performing a test using an injection needle, such as a blood test, as the at least one examination item. The vehicle **30** displays, according to the request data D_q , a text message such as "Do not perform a blood test because the vehicle is about to travel on a rough road" on the display. Alternatively, the vehicle **30**

may output an audio message of similar contents from the speaker. When the time $Tm4$ is X minutes after the current time, the vehicle **30** may display, on the display, a text message such as “Please perform a blood test in X minutes because the vehicle is about to travel on a rough road”. Alternatively, the vehicle **30** may output an audio message of similar contents from the speaker.

(87) According to this example, the examination using the injection needle can be postponed while the vehicle **30** is traveling on a rough road. As a result, safety is improved when the medical examination is performed in the vehicle **30**.

(88) In another example, the request data Dq includes data requesting to postpone performing a test to be performed after the user **11** once stands up, such as a test subsequent to a test performed on the user **11** sitting on a chair, as the at least one examination item. The vehicle **30** displays, according to the request data Dq , a text message such as “Do not let the patient stand up from the chair because the vehicle is about to travel on a rough road” on the display. Alternatively, the vehicle **30** may output an audio message of similar contents from the speaker. When the time $Tm4$ is X minutes after the current time, the vehicle **30** may display, on the display, a text message such as “Please let the patient stand up from the chair in X minutes because the vehicle is about to travel on a rough road”. Alternatively, the vehicle **30** may output an audio message of similar contents from the speaker.

(89) According to this example, it is possible to keep the user **11** from standing up while the vehicle **30** is traveling on a rough road, especially when the user **11** is an elderly person. As a result, safety is improved when the medical examination is performed in the vehicle **30**.

(90) The operations illustrated in FIG. 6 are repeated periodically or whenever a specific event occurs, such as a change in the driving condition of the vehicle **30**.

(91) The process in step **S403** may be omitted. In other words, the request data Dq output in step **S404** need not be added with data indicating the time $Tm4$ when the vehicle **30** arrives at the road meeting the criterion.

(92) In step **S404**, as in step **S104** of FIG. 3, the controller **21** of the control apparatus **20** may output the request data Dq to the terminal apparatus of the medical staff.

(93) As a variation of the present embodiment, the controller **21** of the control apparatus **20** may further acquire route data Dr indicating a route along which the vehicle **30** is to move. The controller **21** may set, based on the acquired route data Dr , implementation order of two or more examination items included in the medical examination to be performed in the vehicle **30**.

(94) For example, the vehicle **30** transmits, as the route data Dr , data indicating a route to a destination, such as a hospital, set in navigation equipment installed in the vehicle **30** to the control apparatus **20** via the communication interface of the vehicle **30**. The controller **21** of the control apparatus **20** acquires the route data Dr by receiving the route data Dr from the vehicle **30** via the communication interface **23**. The controller **21** determines, as with the procedure in step **S402** of FIG. 6, a road surface condition of each road included in the route indicated by the acquired route data Dr . The controller **21** optimizes the implementation order of the two or more examination items included in the medical examination, so that some examination items that may cause danger, such as a test using an injection needle or a test to be performed after the user **11** once stands up, are not performed while the vehicle **30** is traveling on a rough road.

(95) As a further variation, the controller **21** of the control apparatus **20** may acquire evaluation data De indicating an evaluation by the user **11** when the vehicle **30** travels on each road included in the route indicated by the route data Dr . The controller **21** may refer to the acquired evaluation data De to set implementation order of examination items in the next medical examination. For example, suppose that the evaluation by the user **11**, while the vehicle **30** travels on a certain road, is poor such as dissatisfaction because vibration of the vehicle **30** causes more pain by an injection needle. In this case, even if a road surface condition of the road is determined to be good, the controller **21** may optimize the implementation order of the examination items in the next medical examination so that a test using an injection needle is not performed while the vehicle **30** travels on

the road.

(96) The present disclosure is not limited to the embodiment described above. For example, two or more blocks described in the block diagrams may be integrated, or a block may be divided. Instead of executing two or more steps described in the flowcharts in chronological order in accordance with the description, the steps may be executed in parallel or in a different order according to the processing capability of the apparatus that executes each step, or as required. Other modifications can be made without departing from the spirit of the present disclosure.

(97) For example, the control apparatus **20** may be included in the vehicle **30**.

Claims

1. A control apparatus comprising a controller configured to: acquire positioning results obtained by a global navigation satellite system receiver mounted on a vehicle in which a medical examination can be performed or sensing results obtained by at least one sensor mounted on the vehicle selected from the group consisting of a camera, LiDAR and radar; analyze the acquired positioning or sensing results to determine whether a speed of the vehicle exceeds a threshold value; and upon determining that the speed of the vehicle exceeds the threshold value: automatically receive, from an external system, or generate, by analyzing the sensing results, road traffic information including a speed limit, a traffic congestion, or a traffic light timing on a road beyond a current position of the vehicle; predict, based on the automatically received or generated road traffic information, a time when the speed of the vehicle becomes equal to or less than the threshold value; and output, through a display or a speaker in the vehicle, a message requesting a postponement of at least one examination item included in the medical examination.
2. The control apparatus according to claim 1, wherein the at least one examination item includes a test using an injection needle.
3. The control apparatus according to claim 1, wherein the at least one examination item includes a test to be performed after a user who is having the medical examination once stands up.
4. A system comprising: the control apparatus according to claim 1; and the vehicle.
5. A vehicle comprising the control apparatus according to claim 1.
6. A control apparatus comprising a controller configured to: acquire positioning results obtained by a global navigation satellite system receiver mounted on a vehicle in which a medical examination can be performed or sensing results obtained by at least one sensor mounted on the vehicle selected from the group consisting of a camera, LiDAR and radar; analyze the acquired positioning or sensing results to determine whether the vehicle is waiting for a traffic light to change or whether the vehicle is moving, there is a traffic light beyond a current position of the vehicle, and the vehicle is to stop at the traffic light; upon determining that the vehicle is waiting for a traffic light to change, output, through a display or a speaker in the vehicle, a message requesting postponement of at least one examination item included in the medical examination until the vehicle starts; and upon determining that the vehicle is moving, there is a traffic light beyond the current position of the vehicle, and the vehicle is to stop at the traffic light, output, through the display or the speaker in the vehicle, a message requesting a postponement of the at least one examination item until the vehicle stops.
7. The control apparatus according to claim 6, wherein the at least one examination item includes a test using an injection needle.
8. The control apparatus according to claim 6, wherein the at least one examination item includes a test to be performed after a user who is having the medical examination once stands up.
9. A system comprising: the control apparatus according to claim 6; and the vehicle.
10. A vehicle comprising the control apparatus according to claim 6.
11. A control apparatus comprising a controller is configured to: acquire positioning results obtained by a global navigation satellite system receiver mounted on a vehicle in which a medical

examination can be performed or sensing results obtained by a sensor mounted on the vehicle selected from the group consisting of a camera, LiDAR and radar; analyze the acquired positioning or sensing results to determine a road surface condition of a road beyond a current position of the vehicle; and upon determining that the determined road surface condition does not meet a criterion: identify, based on map data or the sensing results, a road that is further ahead of the road beyond the current position of the vehicle and that meets the criterion; automatically receive, from an external system, or generate, by analyzing the sensing results, road traffic information including a speed limit, a traffic congestion, or a traffic light timing on each road between the current position of the vehicle and the identified road that meets the criterion; predict, based on the automatically received or generated road traffic information, a time when the vehicle arrives at the identified road that meets the criterion; and output, through a display or a speaker in the vehicle, a message requesting a postponement of at least one examination item included in the medical examination until the predicted time.

12. The control apparatus according to claim 11, wherein the at least one examination item includes a test using an injection needle.

13. The control apparatus according to claim 11, wherein the at least one examination item includes a test to be performed after a user who is having the medical examination once stands up.

14. A system comprising: the control apparatus according to claim 11; and the vehicle.

15. A vehicle comprising the control apparatus according to claim 11.
